

Linear Control Design 1

Exam Questionnaire F21

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Question 11

Consider a system with transfer function

$$G(s) = \frac{120}{s^3 + 43s^2 + 120s}$$

and Bode plot as shown in the following figure. A P-controller K_P is applied to the system

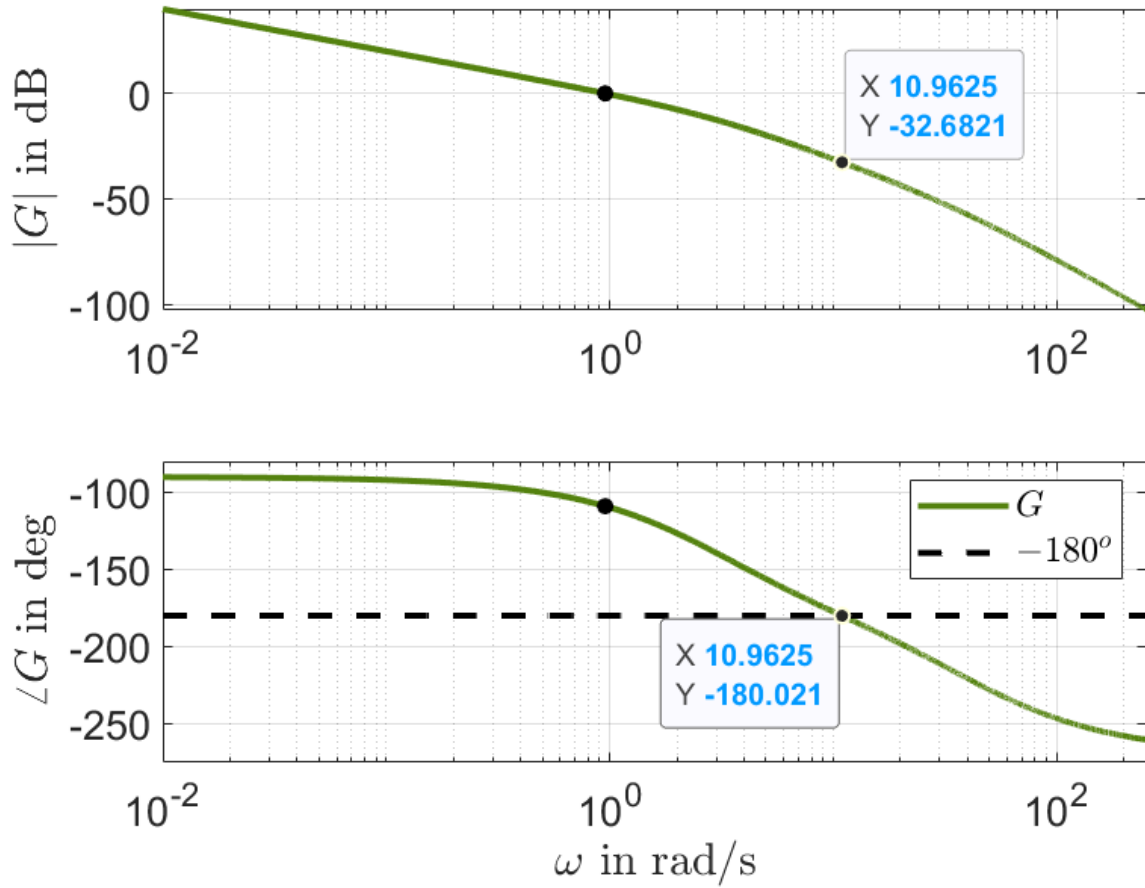


Figure 1: Bode plot

and the loop is closed with unit feedback. The closed-loop system is stable for:

- a) $K_P = -1$
- b) $K_P = 25$
- c) $K_P = 71$
- d) $K_P = -32$
- e) $K_P = 45$

Question 12

Consider the system described by the following differential equation

$$y^{(4)}(t) + 9y^{(3)}(t) + 20\ddot{y}(t) = 71u(t)$$

where $\ddot{y}(t) = \frac{d^2y}{dt^2}$ and $y^{(n)}(t) = \frac{d^ny}{dt^n}$ denotes the n^{th} time derivative of $y(t)$. Which of the following statements is the correct one?

- a) The system is stable.
- b) The system is nonlinear.
- c) The system has two zeros.
- d) The system is unstable.
- e) The steady state (DC) gain of G is 3.55.

Question 13

Consider a system with open-loop transfer function

$$G(s) = \frac{s+1}{s^2+43s-90}$$

and Nyquist plot as shown in the following figure (Fig. 2). The Nyquist plot intersects the real axis along the counter-clockwise direction at the point $(-0.0111, 0)$. A P-controller K_P is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

- a) $K_P = 100$
- b) $0.02 < K_P < 1$
- c) $0 < K_P < 80$
- d) $K_P < -0.0111$
- e) $0 < K_P < 0.02$

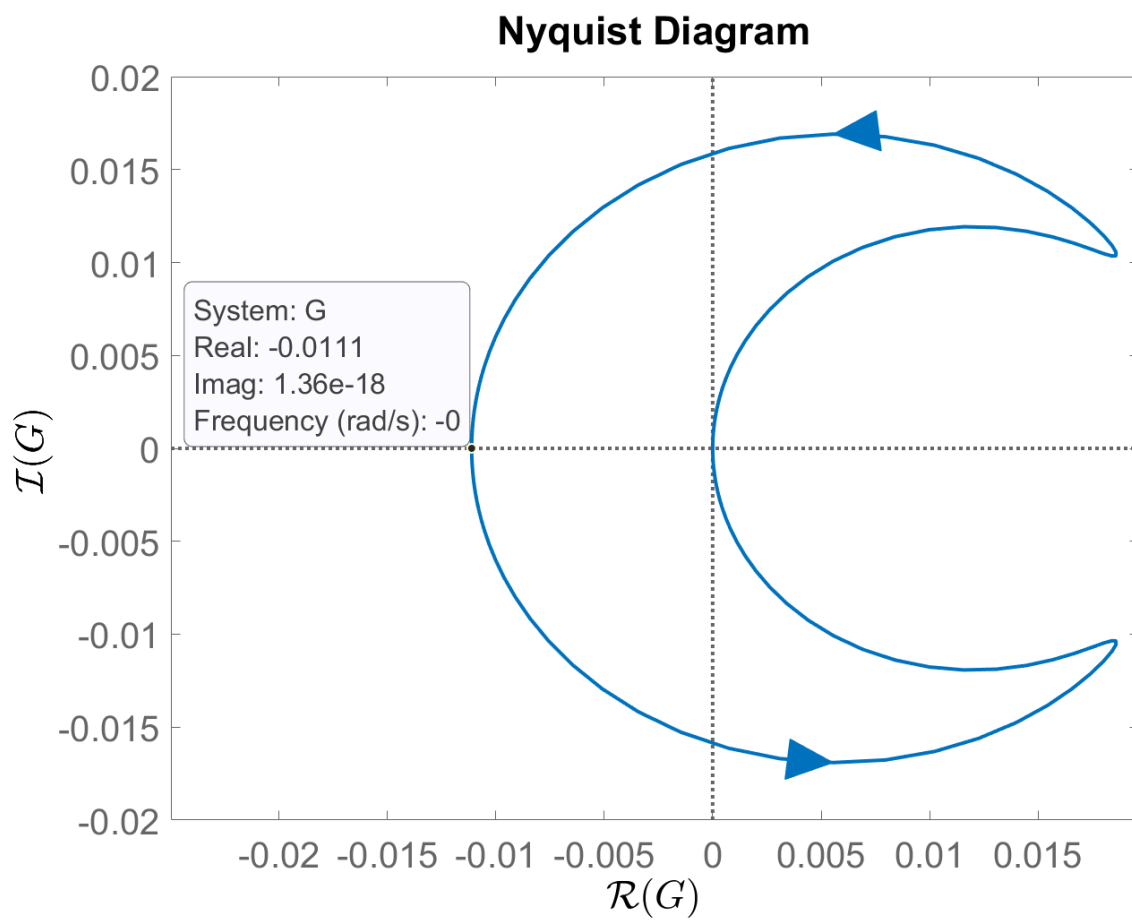


Figure 2: Nyquist plot

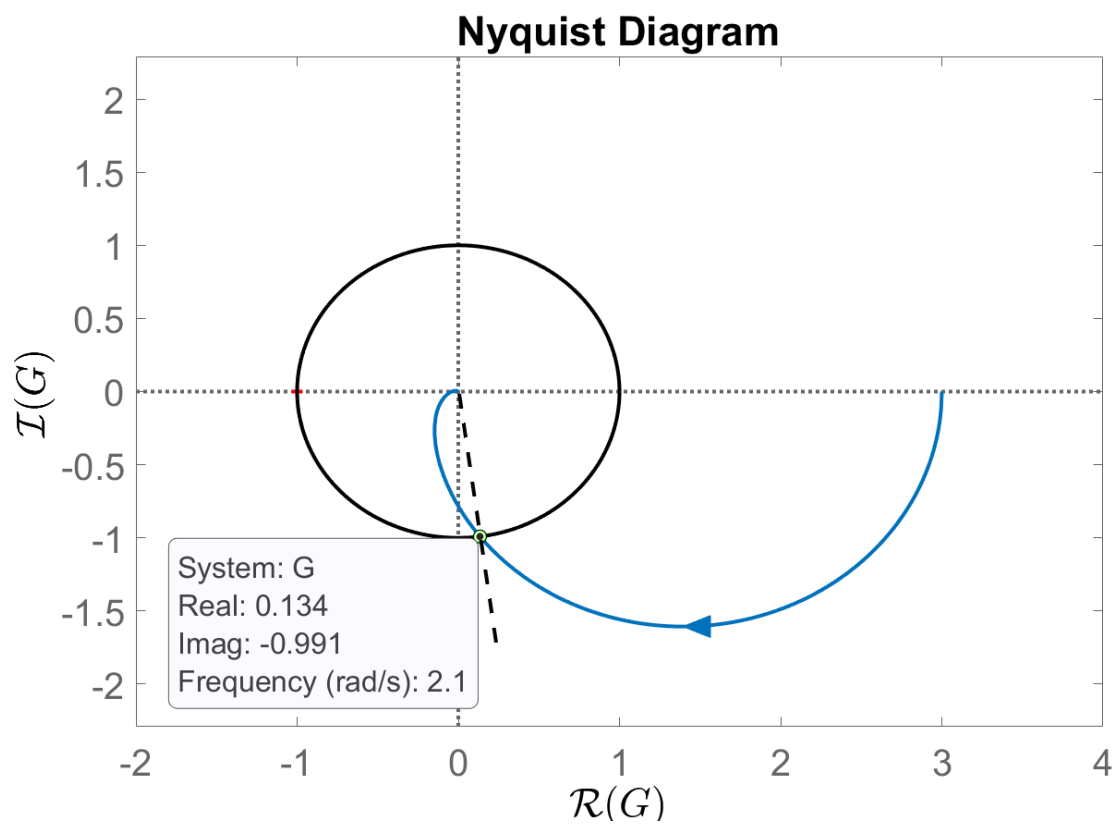


Figure 3: Nyquist plot

Question 14

Consider a system with transfer function

$$G(s) = \frac{1200}{s^3 + 54s^2 + 570s + 400}$$

and Nyquist plot (only positive frequencies) as shown in the following figure (Fig. 3). The Nyquist plot intersects the unit circle along the clockwise direction at the point $(0.134, -0.99)$. The phase margin of the system is approximately

- a) $\gamma_M = 0.1354^\circ$
- b) $\gamma_M = 1.705^\circ$
- c) $\gamma_M = -1.44^\circ$
- d) $\gamma_M = 82.29^\circ$
- e) $\gamma_M = 97.7^\circ$

Question 15

Consider a second-order type-0 system with two stable real poles and no zeros. A P-controller with $K_P > 1$ is applied to the system. Which of the following statements is the correct one?

- a) Increasing K_P will shift the phase curve in the open-loop Bode plot upwards.
- b) Increasing K_P will increase the crossover-frequency of the open-loop system.
- c) Increasing K_P too much can theoretically make the closed-loop system unstable.
- d) After a finite value of K_P (i.e. $K_P < \infty$), the steady state error in the output of the closed-loop system due to a step change in the reference signal will be zero.
- e) The open-loop system can have multiple crossover-frequencies.

Question 16

Consider a system with transfer function

$$G(s) = \frac{1224}{s^3 + 30s^2 + 257s + 612}$$

A P-controller with $K_P = 2$ is applied to the system but it is placed in the feedback branch as in the following figure (Fig. 4). The steady state error $e(s)$ in the closed-loop system due to a unit step change in the reference signal is:

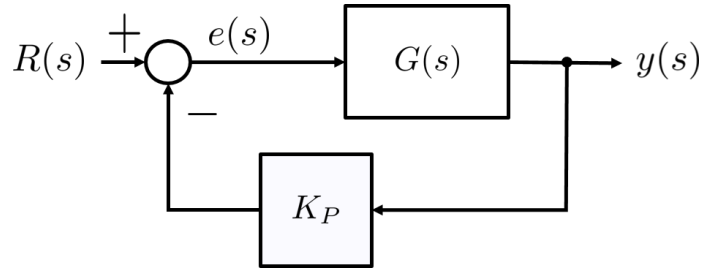


Figure 4: Closed-loop system.

- a) $e_{SS} = 0$
- b) $e_{SS} = 2$
- c) $e_{SS} = 0.2$
- d) $e_{SS} = 0.4$
- e) $e_{SS} = 0.8$

Question 17

Consider the system described by the following transfer function and Bode plot (Fig. 5)

$$G(s) = \frac{5(s+7)(s+12)}{(s+4)(s+9)(s+17)(s+5)(s+70)}$$

A P-Lead-**Lag** controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$

with $\alpha = 0.2$ needs to be designed for the system. The new crossover frequency (after applying the controller) is selected as $\omega_c = 15$ rad/s and the phase contribution of the Lag part at that frequency is given in radians by

$$\phi_L = \arctan \left(N_i \frac{1 - \beta}{1 + \beta N_i^2} \right),$$

where $N_i = \omega_c \tau_i = 3$. If a phase margin $\gamma_M = 70^\circ$ is desired for the open-loop system $C(s)G(s)$, what is the value of the parameter β of the Lag-part?

- a) $\beta = 0$
- b) $\beta = 2$
- c) $\beta = -1.5$
- d) $\beta = 0.3$
- e) $\beta = 0.62$

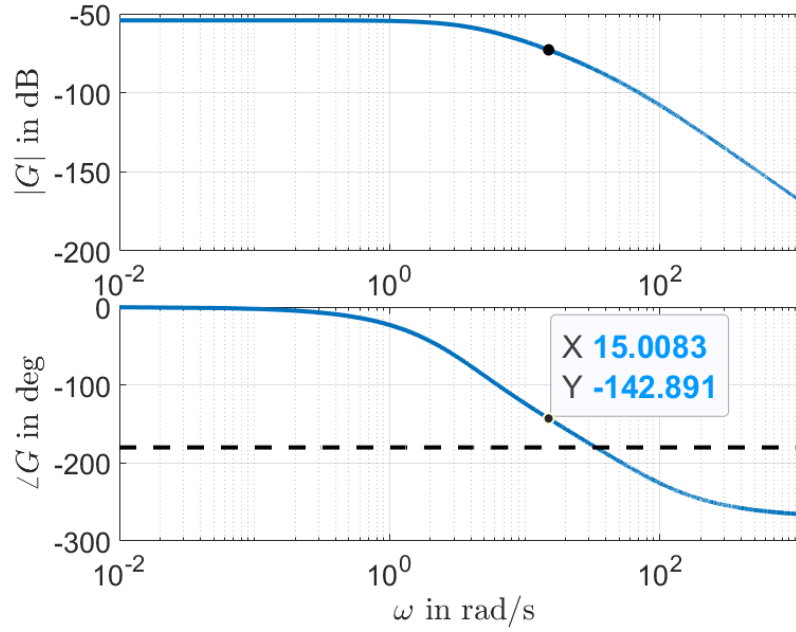


Figure 5: Bode plot

Question 18

A system with transfer function

$$G(s) = \frac{0.7(s + 0.5)}{(5s + 1)(s^2 + 0.2s + 0.6)(0.01s + 1)}$$

has the following Bode plot (Fig. 6)

A PI-Lead controller

$$C(s) = K_P \frac{\tau_i s + 1}{\tau_i s} \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 45^\circ$ and crossover frequency $\omega_c = 10$ rad/s. If $N_i = \omega_c \tau_i = 8$, a good selection for α and K_P is:

- a) $\alpha = 0.5$ and $K_P = 500$.
- b) $\alpha = 2$ and $K_P = 1000$.
- c) $\alpha = 0.7$ and $K_P = 591$.
- d) $\alpha = 0.08$ and $K_P = 200$.
- e) $\alpha = 0.07$ and $K_P = -154$.

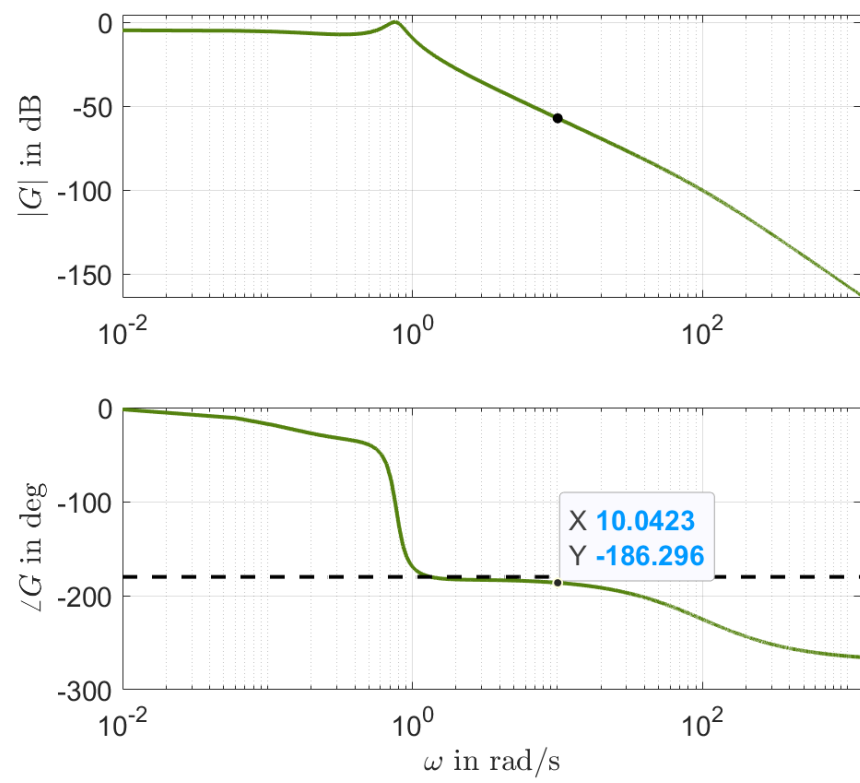


Figure 6: Bode plot

Question 19

Consider the system described by the following block diagram where $G_1(s)$ and $G_2(s)$ are type-0 systems.

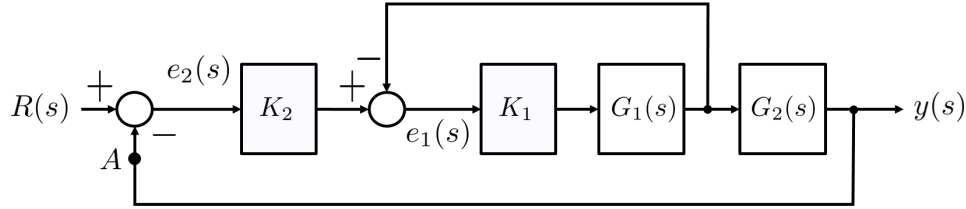


Figure 7: Block diagram.

The magnitude Bode plot of $G_2(s)$ is shown below:

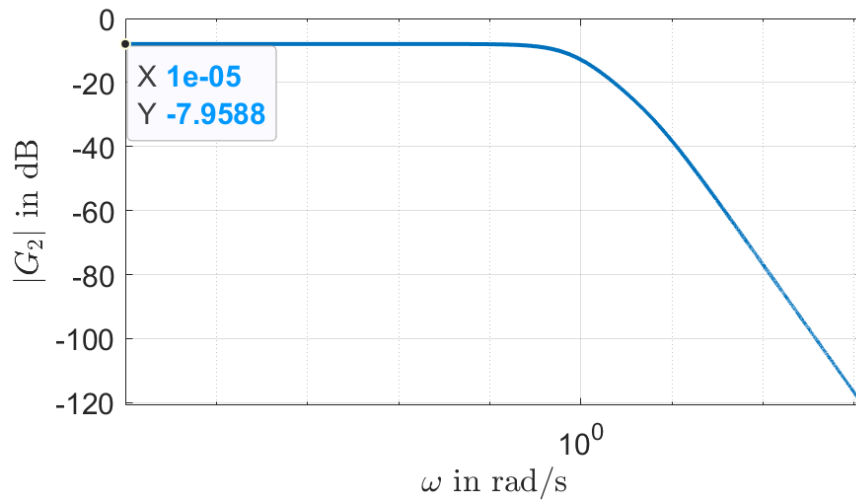


Figure 8: Magnitude Bode plot.

When the outer loop is opened at point A , a step change in $R(s)$ equal to $\frac{1}{K_2}$ causes a steady state error $e_1(0) = 0.4$. What should be the value of K_2 , such that a unit step change in the reference $R(s)$, when the loop is closed, causes a steady state error $e_2(0) = 0.05$?

- a) $K_2 = 118.75$.
- b) $K_2 = -3.98$.
- c) $K_2 = -5.97$.
- d) $K_2 = 91.25$.
- e) $K_2 = 79.17$.

Question 20

Consider the system described by the following block diagram,

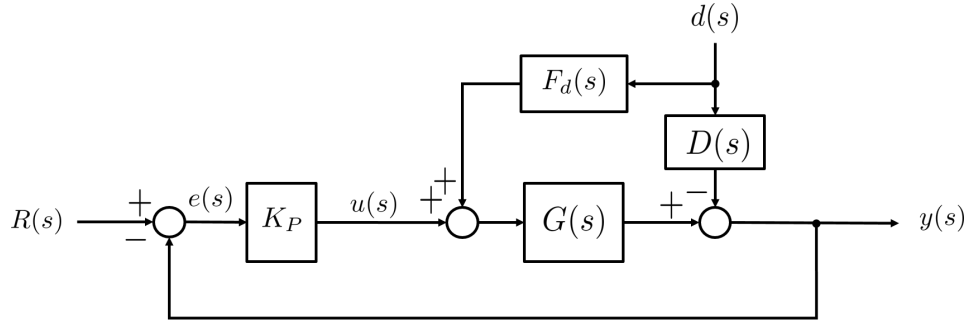


Figure 9: Block diagram.

where $K_P = 1$, $d(s)$ is an external disturbance and the transfer functions $G(s)$ and $D(s)$ are defined as

$$G(s) = \frac{21(0.5s + 1)}{s^2 + 4s + 21}$$

$$D(s) = \frac{1}{0.01s + 1}$$

If $F_d(s)$ is an unfiltered dynamic feed-forward controller designed to compensate for the disturbance, then the sensitivity function related to the disturbance $d(s)$ is:

- a) $G_{yd}(s) = 0$.
- b) $G_{yd}(s) = \frac{100s^2 + 400s + 2100}{s^3 + 114.5s^2 + 1492s + 4200}$.
- c) $G_{yd}(s) = -\frac{100s^2 + 400s + 2100}{s^3 + 114.5s^2 + 1492s + 4200}$.
- d) $G_{yd}(s) = \frac{100}{s + 1}$.
- e) $G_{yd}(s) = \frac{10.5s^2 + 1071s + 2100}{s^3 + 104s^2 + 1471s + 4200}$.